

Caleb Kang

(470) 357-1857 | cjkang2@illinois.edu | linkedin.com/in/calebjkang | github.com/CalebK

EDUCATION

University of Illinois Urbana-Champaign

Urbana, IL

Bachelor of Science in Computer Science, Grainger College of Engineering

Aug. 2026 – May 2028

- Guaranteed admission via Grainger Engineering Pathways program

William Rainey Harper College

Palatine, IL

Associate in Engineering Science, GPA: 4.0/4.0

Aug. 2024 – May 2026

- **Awards:** Motorola Solutions Engineering Pathways Scholarship, Full-Ride Scholarship, Honors Society, President's List
- **Activities:** Computer Science Club, Society of Engineers
- **Relevant Coursework:** Computer Science I & II, Data Structures & Algorithms, Linear Algebra, Discrete Mathematics, Differential Equations

EXPERIENCE

Software Development Engineer Intern

May 2026 – Aug. 2026

Amazon

Bellevue, WA

- Incoming SDE Intern on the Amazon Fintech team.

Software Engineer Intern

Jun. 2025 – Aug. 2025

Fermi National Accelerator Laboratory

Batavia, IL

- Engineered a Python state machine to automate ASIC chip testing for the Deep Underground Neutrino Experiment (DUNE), scaling infrastructure across six national sites to validate $\approx 300,000$ chips
- Designed a comprehensive UML state diagram to map logical flow and transition states, ensuring robust error handling prior to implementation
- Authored a technical research report and delivered findings to scientists and academics via poster sessions and virtual reviews, detailing testing methodologies and system architecture

Research Assistant

Oct. 2024 – May 2025

Biola University (Remote)

La Mirada, CA

- Validated multimodal authentication models by fine-tuning and deploying CLIP, Whisper, and Sortformer architectures
- Executed NVIDIA NeMo model performance testing in Google Colab using PyTorch to assess viability for security applications

PROJECTS

C++ Farming Simulator | C++, OOP, Catch2

Oct. 2025 – Dec. 2025

- Architected a terminal-based simulation game using object-oriented principles (inheritance, polymorphism) to model complex crop lifecycles and interactions
- Implemented a dynamic 2D grid system using vectors of pointers, utilizing manual memory management to efficiently handle object allocation and deallocation during gameplay state changes
- Engineered a comprehensive suite of unit tests using the Catch2 framework to validate game logic, ensuring stability across edge cases in player movement and resource calculation

Duckiebot Robotics Initiative | Linux, Docker, Raspberry Pi, NVIDIA Jetson

Dec. 2024 – Jan. 2025

- Spearheaded a 4-person team in the development of a robotic rover, managing project roadmap and coordinating weekly technical sprints
- Configured embedded Linux (Ubuntu) environments on Raspberry Pi and NVIDIA Jetson hardware, troubleshooting kernel-level networking and terminal configurations
- Deployed the Duckietown robotics framework using Docker containers, successfully establishing command-and-control connectivity with the rover

TECHNICAL SKILLS

Languages: Java, Python, C++, TypeScript, JavaScript, HTML/CSS

AI & Machine Learning: PyTorch, Generative AI, LLMs, CLIP, Whisper, Sortformer, NVIDIA NeMo

Developer Tools: Git, GitHub, Docker, Linux, React.js

Cloud: Google Cloud Platform (GCP)